UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

End of Term 1 Exam Coverage	هيكل اختبار نهاية الفصل الدراسي الأول
Grade 12 Advanced - Chemistry	**الصف الثاني عشر - متقدم - الكيمياء**
Academic Year 2026-2027	العام الدراسي 2027-2026
Inspire	انسباير

عدد الأسئلة Number of questions	درجة السؤال Marks per question	الدرجة القصوى الممكنة Maximum overall grade
30	**(3,4)**	**100**

مدة الاختبار Exam duration	طريقة التطبيق Mode of implementation	الآلة الحاسبة Calculator
120 دقيقة **120 minutes**	**إلكتروني** **Electronic**	**يُسمح باستخدام الآلة الحاسبة العلمية** **The use of a scientific calculator is allowed**

⚠ قد تظهر الأسئلة بترتيب مختلف في الاختبار الفعلي.

⚠ Questions might appear in a different order in the actual exam.
NO. of pages from inspire chemistry -UAE- edition grade 11-12 advanced Volume 2

نوع الأسئلة Question type	Q	Learning outcome	Inspire (English)	Page
أسئلة الاختيار من متعدد Multiple Choice Questions	Q1	Distinguish between oxidation and reduction in terms of loss and gain of electrons, oxygen and hydrogen	figures 1,2+ textbook	3,4
	Q2	Define oxidation number	textbook	5
	Q3	Define oxidizing agent and reducing agent in a redox reaction	table 2	6
	Q4	Identify oxidizing agent and reducing agent in a redox reaction	textbook +example problem 1 + practice problems	5,6,8

نوع الأسئلة Question type	Q	Learning outcome	Inspire (English)	Page
أسئلة اختيار من متعدد Multiple Choice Questions	Q5	Identify oxidizing agent and reducing agent in a redox reaction	textbook+ example problem1 + practice problems	5,6,8
	Q6	Distinguish between oxidation and reduction in terms of change in oxidation number	textbook +example problem1+ practice problems	5,6,8
	Q7	Assign oxidation number to atoms, ions and compounds according to a set of rules	textbook +example problem 2 + practice problems+table3	9,10
	Q8	Assign oxidation number to atoms, ions and compounds according to a set of rules	textbook+ example problem2 + practice problems+table3	9,10
	Q9	Determines the oxidation number of elements and compounds with more complex formulas.	textbook+ example problem2+ practice problems+table3	9,10
	Q10	Distinguish between oxidation and reduction in terms of change in oxidation number	check your progress and textbook	11
	Q11	Distinguish between oxidation and reduction in terms of change in oxidation number	check your progress and textbook	11
	Q12	Describe the steps for balancing redox reactions, in acidic medium, by the half-reaction method	table 6	17
	Q13	Write oxidation-half reaction and reduction-half reaction for a redox reaction	figure 8+ textbook+ table 5	16,17
	Q14	Balance redox reaction in acidic medium using half-reaction method	textbook +example problem 5 + practice problems and check your progress	16,17,18, 19
	Q15	Balance redox reaction in acidic medium using half-reaction method	textbook +example problem 5 + practice problems	16,17,18
	Q16	Describe the steps for balancing redox reactions, in basic medium, by the half-reaction method	textbook +example problem 5 + practice problems and check your progress	16,17,18, 19
	Q17	Balance redox reaction in basic medium using half-reaction method	textbook +example problem 5 + practice problems and check your progress	16,17,18, 19
	Q18	Describe an electrochemical cell while specifying its types	figures 1,2+ textbook	25,26

نوع الأسئلة Question type	Q	Learning outcome	Inspire (English)	Page
أسئلة اختيار من متعدد Multiple Choice Questions	Q19	Identify components of a voltaic or galvanic cell (anode, cathode, salt bridge or porous barrier, wires, electrolyte compartments); while explaining the role of each component, when does the reaction start and determining the direction of electron and current flow	figures 1,2+ textbook	25,26
	Q20	Write the oxidation and reduction half-reactions occurring at cathode and anode for a voltaic cell	textbook+fiqure 6	27,31
	Q21	Define the reduction potential and standard electrode potential (E°)	table 1+ textbook	29,30
	Q22	Describe standard hydrogen electrode (SHE), while identifying the importance of its E° value and writing the half-cell reactions of the two possible reactions that could occur at the hydrogen electrode	figures 5,6+ textbook	29,31
	Q23	Use the half-cell standard reduction potentials to calculate the electrochemical cell standard potential, while determining whether the redox reactions are spontaneous or non-spontaneous	textbook+ example problem1 + practice problems+table3	32,33,34
	Q24	Write the cell notation and the overall chemical equation for a redox reaction occurring in a voltaic cell	textbook+ example problem1 + practice problems+table3	31,32,33, 34
	Q25	Use the half-cell standard reduction potentials to calculate the electrochemical cell standard potential, while determining whether the redox reactions are spontaneous or non-spontaneous	textbook+ practice problems+check your progress	34,35
	Q26	Compare between electrolytic cell and voltaic cell in terms of identifying where will reduction and oxidation processes take place, anode, cathode, direction of electron flow and current flow and spontaneity of the reaction occurring	figure 18+ textbook	47
	Q27	Describe how a spontaneous redox reaction of an electrochemical cell can be reversed	figure 18+ textbook	47,48
	Q28	Write half-cell reactions while identifying the products of electrolysis of molten ionic compounds	figure 19+ textbook	48
	Q29	Write half-cell reactions while identifying the products of electrolysis of aqueous ionic compounds	figure 20+ textbook	49
	Q30	Define electroplating while describing how it works, identifying anode, cathode and electrolyte needed for an electrolytic cell in which a selected metal is to be plated on an object (car or spoon,... etc)	figure 22+ textbook	51,52